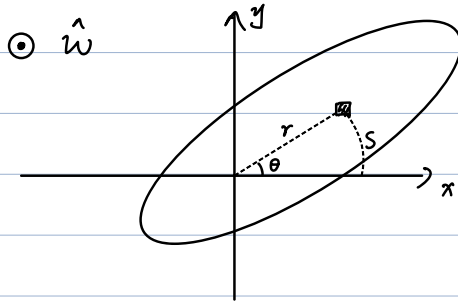


Point particle $\rightarrow$ extended rigid object
 $\downarrow$
non-deformable

New degree of freedom: Rotation

Today: Rotation of rigid body about a fixed axis

In the plane perpendicular to the rotation axis:



Polar coordinates: (r, θ)

arc length: $s = r\theta$

A few definitions on angular motion:

- Angular displacement:

$$\Delta\theta = \theta_f - \theta_i$$

- Average angular speed:

$$w_{ave} \equiv \frac{\Delta\theta}{\Delta t}$$

- Instantaneous angular speed:

$$w \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

- unit: rad/s , or s^{-1}

- direction: "+" for counter clockwise

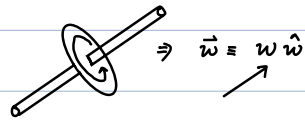
- Average angular acceleration

$$\alpha_{ave} \equiv \frac{\Delta w}{\Delta t}$$

- Instantaneous angular acceleration

$$\alpha \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} = \frac{dw}{dt}$$

- Right-hand rule



(then the counter clock wise direction is defined)

- Rotational Kinematics

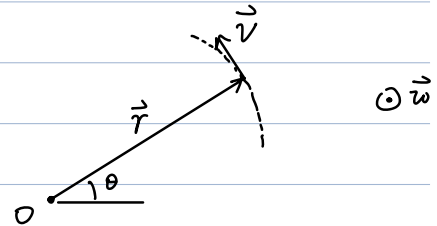
- Constant angular acceleration

Rotation	Linear Motion
$\omega_f = \omega_i + \alpha t$	$v_f = v_i + a t$
$\theta_f = \theta_i + \omega_i t + \frac{1}{2} \alpha t^2$	$x_f = x_i + v_i t + \frac{1}{2} a t^2$
$= \frac{\omega_f^2 - \omega_i^2}{2\alpha}$	$= x_i + \frac{v_f^2 - v_i^2}{2a}$

$$x \leftrightarrow \theta, \quad v \leftrightarrow \omega, \quad a \leftrightarrow \alpha$$

- Relation between $\vec{v}$ and $\vec{\omega}$

$$\begin{aligned} v &= \frac{ds}{dt} = \frac{d(ro)}{dt} \\ &= r \frac{d\theta}{dt} \\ &= \omega r \end{aligned}$$



In vector form:

$$\vec{v} = \vec{\omega} \times \vec{r}$$

(more discussion on "x" next class)

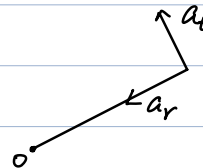
- Relation between $\vec{a}$ and $\vec{\omega}$

- separate $\vec{a} = \vec{a}_t + \vec{a}_r$

- $a_t = \frac{dv}{dt} = r \frac{d\omega}{dt} = \alpha r$

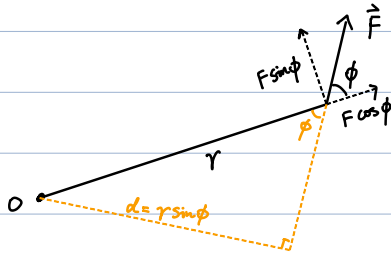
$$a_r = \frac{v^2}{r} = \omega^2 r$$

$$\begin{aligned} \Rightarrow a &= \sqrt{a_t^2 + a_r^2} = \sqrt{\alpha^2 r^2 + \omega^4 r^2} \\ &= \sqrt{\alpha^2 + \omega^4} r \end{aligned}$$



- Torque (力矩)

$$\tau = (F \sin \phi) r = Fd$$



d : moment arm of $\vec{F}$ (力臂)

Notes:

1. Torque defined ONLY WHEN a reference axis is defined.

2. Total torque from multiple forces:

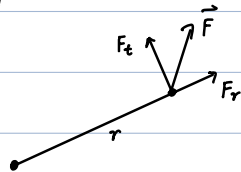
$$\tau_{\text{total}} = \sum_i \tau_i$$

3. Vector form:

$$\vec{\tau} = \vec{r} \times \vec{F}$$

• Torque and angular acceleration

— point particle



$$F_t = m a_t$$

$$\Rightarrow \tau = F_t r = m a_t r = (m r^2) \alpha$$

$\downarrow$
 I

Moment of inertia (转动惯量)

$$I \equiv m r^2$$

$$\Rightarrow \tau = I \alpha \quad (\text{analog of } F = ma)$$

— Generalize to rigid body

$$dF_t = a_t dm$$

$$\Rightarrow d\tau = r dF_t = r a_t dm = r^2 \alpha dm$$

③

$$\Rightarrow \tau_{\text{total}} = \int d\tau = \left(\int r^2 dm \right) \alpha$$

Again, define

$$I = \int r^2 dm$$

$$\Rightarrow \boxed{\tau_{\text{total}} = I \alpha}$$

Notes:

1. same $\vec{w}$, $\vec{a}$ at any point on the rigid body.

($\vec{v}$, $\vec{a}$ can be different)

2. $\{ \vec{\tau}_{\text{total}}, \vec{w}, \vec{a} \}$ characterize the angular motion of rigid body.

(Up to now: $\tau_{\text{total}} = I \alpha$, analog of $F = ma$)

(below: energy and momentum)

• Rotational kinetic energy:

$$\begin{aligned} K_R &= \int dK = \int \frac{1}{2} v^2 dm \\ &= \int \frac{1}{2} w^2 r^2 dm \\ &= \int \frac{1}{2} w^2 dI \end{aligned}$$

$$\Rightarrow K_R = \frac{1}{2} I w^2$$

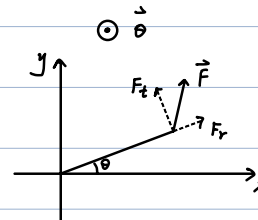
Notes:

1. for fixed axis, K_R just the regular kinetic energy.

2. If axis is moving, then K_R is only the "rotational" part of kinetic energy.

• Work

$$\begin{aligned} dW &= \vec{F} \cdot d\vec{s} \\ &= F_t \cdot r d\theta \\ &= \tau d\theta \end{aligned}$$



vector form:

$$dW = \vec{\tau} \cdot d\vec{\theta} \quad (\vec{\tau} = \vec{r} \times \vec{F})$$

Note:

F_r does no work

- Instantaneous power

$$P = \frac{dW}{dt} = \frac{\tau d\theta}{dt} = \tau \omega$$

- Work - kinetic energy theorem

$$\tau_{\text{total}} = I \alpha$$

$$\Rightarrow \tau_{\text{total}} = I \frac{d\omega}{dt}$$

$$\Rightarrow \tau_{\text{total}} d\theta = I \frac{d\omega}{dt} d\theta$$

$$= I \omega d\omega$$

$$\Rightarrow W_{\text{total}} = \int \tau_{\text{total}} d\theta = \int_{\omega_i}^{\omega_f} I \omega d\omega$$

$$\Rightarrow \boxed{W_{\text{total}} = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2}$$

- Short summary:

Rotation	Linear Motion
$\omega = \frac{d\theta}{dt}$	$v = \frac{dx}{dt}$
$\alpha = \frac{d\omega}{dt}$	$a = \frac{dv}{dt}$
$\tau_{\text{total}} = I \alpha$	$F = ma$
$W = \int_{\theta_i}^{\theta_f} \tau d\theta$	$W = \int_{x_i}^{x_f} F dx$
$K_R = \frac{1}{2} I \omega^2$	$K = \frac{1}{2} m v^2$
$P = \tau \omega$	$P = F v$
$W = K_f - K_i$	$W = K_f - K_i$

Notes:

- Not discovering new laws, but simply deriving motions for extended objects.
- Analogy between I and m
- m is intrinsic; I depends on choice of reference axis.
- Analogy of Impulse - momentum theorem in coming classes.

Example.

1° $\vec{\omega} \parallel \hat{y}$

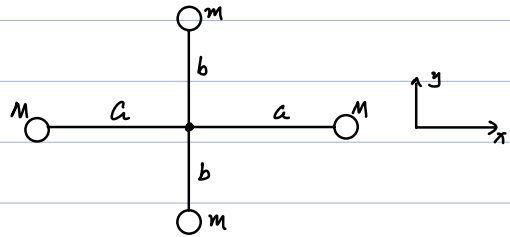
$$I = Ma^2 + Ma^2 = 2Ma^2$$

2° $\vec{\omega} \parallel \hat{x}$

$$I = 2mb^2$$

3° $\vec{\omega} \parallel \hat{z}$

$$I = 2Ma^2 + 2mb^2$$



Example. Uniform rod

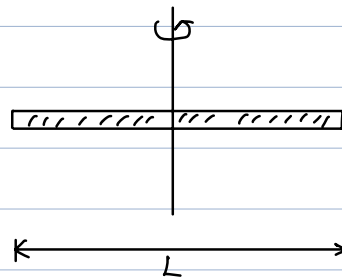
mass in unit length: $\lambda = M/L$

1° axis at center

$$\begin{aligned} I &= \int_{-\frac{L}{2}}^{\frac{L}{2}} x^2 \lambda dx \\ &= \frac{1}{3} \frac{M}{L} x^3 \Big|_{-\frac{L}{2}}^{\frac{L}{2}} \\ &= \frac{1}{12} ML^2 \end{aligned}$$

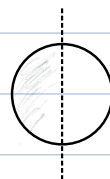
2° axis at end

$$\begin{aligned} I &= \int_0^L x^2 \lambda dx \\ &= \frac{1}{3} \frac{M}{L} x^3 \Big|_0^L \\ &= \frac{1}{3} ML^2 \end{aligned}$$

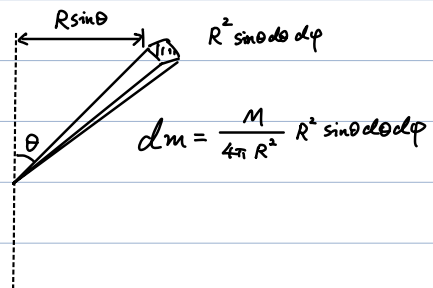


Example. Spherical shell, axis through center.

mass per area: $\frac{M}{4\pi R^2}$



$$\begin{aligned} I &= \int r^2 dm \\ &= \frac{M}{4\pi} \int_0^{2\pi} d\varphi \int_0^\pi \sin\theta d\theta (R \sin\theta)^2 \\ &= \frac{M}{4\pi} \cdot 2\pi R^2 \int_0^\pi \sin\theta (\cos^2\theta - 1) d\theta \\ &= \frac{M}{2} R^2 \left(\frac{1}{3} y^3 - y \right) \Big|_1^{-1} \\ &= \frac{M}{2} R^2 \left(-\frac{1}{3} + 1 - \frac{1}{3} + 1 \right) \\ &= \frac{2}{3} MR^2 \end{aligned}$$



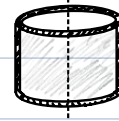
Other examples (proof omitted)

- spherical ball

$$I = \frac{2}{5} MR^2$$

- cylindrical shell

$$I = MR^2$$

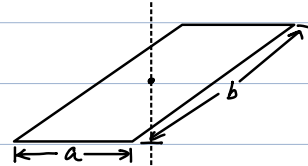


- solid cylinder

$$I = \frac{1}{2} MR^2$$

- rectangular plate

$$I = \frac{1}{12} M(a^2 + b^2)$$



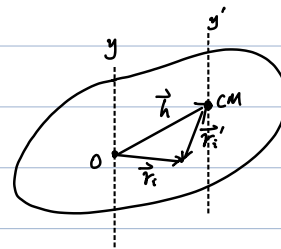
• Parallel axis theorem

$$I = I_{cm} + Mh^2$$

h : displacement of axis from CM

Proof:

$$\begin{aligned} I &= \sum_i m_i r_i^2 \\ &= \sum_i m_i (\vec{r}_i' + \vec{h})^2 \\ &= \sum_i m_i (r_i'^2 + 2\vec{r}_i' \cdot \vec{h} + h^2) \\ &= \underbrace{\sum_i m_i r_i'^2}_{I_{cm}} + \underbrace{(\sum_i m_i) h^2}_M + \underbrace{2(\sum_i m_i \vec{r}_i') \cdot \vec{h}}_{0 \text{ when } y' \text{ through CM}} \end{aligned}$$



Note:

$$\begin{aligned} \vec{r}_{cm} &= \frac{\sum_i m_i \vec{r}_i}{M} \Rightarrow \sum_i m_i \vec{r}_i - (\sum_i m_i) \vec{r}_{cm} = 0 \\ &\Rightarrow \sum_i m_i (\vec{r}_i - \vec{r}_{cm}) = 0 \end{aligned}$$

Example: uniform rod

$$I_{cm} = \frac{1}{12} ML^2$$

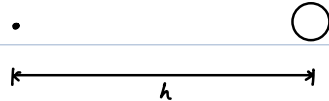
$$\begin{aligned} \Rightarrow I_{end} &= I_{cm} + M\left(\frac{L}{2}\right)^2 \\ &= \frac{1}{3} ML^2 \end{aligned}$$

Example: spherical shell

$$I_{cm} = \frac{2}{3}MR^2$$

$$\Rightarrow I_{end} = I_{cm} + MR^2 = \frac{5}{3}MR^2$$

Example: spherical shell around a far-away axis



$$\Rightarrow I_{far} = I_{cm} + Mh^2$$

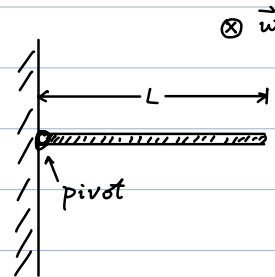
$$= M(h^2 + \frac{2}{3}R^2)$$

$$h \gg R \Rightarrow Mh^2 \rightarrow \text{point-particle like}$$

Example. Rotating rod.

Q: initial angular acceleration?

initial linear acceleration of the right end?



$$\tau_{total} = \int_0^L \frac{M}{L} dx \cdot g \cdot x$$

$$= \frac{M}{L} g \left[\frac{1}{2} x^2 \right]_0^L = \frac{1}{2} MgL$$

$$I = \frac{1}{3}ML^2 \quad (\text{shown earlier})$$

$$\tau_{total} = I\alpha$$

$$\Rightarrow \alpha = \frac{\frac{1}{2}MgL}{\frac{1}{3}ML^2} = \frac{3}{2} \frac{g}{L}$$

$$\Rightarrow a_t = \alpha L = \frac{3}{2}g > g$$